

Import Libraries and Load Dataset

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
```

```
In [2]: data = pd.read_csv('messy_customer_sales_data.csv')
data.head()
```

Out[2]:

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date
0	CUST4371	Paul Wilson	m	52.0	KOLKATA	2025-06-26	2025-05-17
1	CUST5957	Jason Thomas	M	51.0 years	NaN	2021-02-17	2025-07-22
2	CUST3754	Brittney Martinez	F	62.0	hyderabad	2023-11-05	2024-12-08
3	CUST2934	Brenda Pierce	FEMALE	40.0	hyderabad	2022-03-13	2025-10-02
4	CUST5683	Matthew Carroll	f	41.0	CHENNAI	2024-04-05	2024-12-15

```
In [3]: data.shape
```

Out[3]: (10200, 12)

```
In [4]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10200 entries, 0 to 10199
Data columns (total 12 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Customer_ID         9177 non-null   object
1   Name                10200 non-null  object
2   Gender              9174 non-null   object
3   Age                 9249 non-null   object
4   City                9184 non-null   object
5   Signup_Date         10200 non-null  object
6   Last_Purchase_Date  9188 non-null   object
7   Purchase_Amount     9179 non-null   float64
8   Feedback_Score      9177 non-null   float64
9   Email               10200 non-null  object
10  Phone_Number        10200 non-null  int64
11  Country             9468 non-null   object
dtypes: float64(2), int64(1), object(9)
memory usage: 956.4+ KB
```

```
In [5]: data.describe().round()
```

Out[5]:

	Purchase_Amount	Feedback_Score	Phone_Number
count	9179.0	9177.0	1.020000e+04
mean	29090.0	5.0	4.979974e+09
std	208697.0	3.0	2.902593e+09
min	-500.0	1.0	9.208990e+05
25%	12295.0	3.0	2.449157e+09
50%	24330.0	5.0	4.988639e+09
75%	37130.0	8.0	7.510448e+09
max	999999.0	10.0	9.994402e+09

Check Missing Values

In [6]: `data.isnull().sum()`

Out[6]:

Customer_ID	1023
Name	0
Gender	1026
Age	951
City	1016
Signup_Date	0
Last_Purchase_Date	1012
Purchase_Amount	1021
Feedback_Score	1023
Email	0
Phone_Number	0
Country	732

dtype: int64

Remove Duplicate Records

In [7]: `data.duplicated().sum()`

Out[7]: `np.int64(15)`

In [8]: `data = data.drop_duplicates()`

In [9]: `data.shape`

Out[9]: `(10185, 12)`

In [10]: `data.head()`

Out[10]:

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date
0	CUST4371	Paul Wilson	m	52.0	KOLKATA	2025-06-26	2025-05-17
1	CUST5957	Jason Thomas	M	51.0 years	NaN	2021-02-17	2025-07-22
2	CUST3754	Brittney Martinez	F	62.0	hyderabad	2023-11-05	2024-12-08
3	CUST2934	Brenda Pierce	FEMALE	40.0	hyderabad	2022-03-13	2025-10-02
4	CUST5683	Matthew Carroll	f	41.0	CHENNAI	2024-04-05	2024-12-15

Clean Customer ID

```
In [11]: data['Customer_ID'].isnull().sum()
```

```
Out[11]: np.int64(1022)
```

```
In [12]: # fill missing values
data['Customer_ID'] = data['Customer_ID'].fillna('Unknown')
```

Clean Gender Column

```
In [13]: print(data['Gender'].unique())
```

```
['m ' 'M' 'F' 'FEMALE' 'f ' 'male' 'MALE' 'female' nan]
```

```
In [14]: data['Gender'] = data['Gender'].str.strip().str.upper() #convert into uppercas
```

```
In [15]: print(data['Gender'].head(5))
```

```
0      M
1      M
2      F
3  FEMALE
4      F
Name: Gender, dtype: object
```

```
In [16]: # replace values
data['Gender'] = data['Gender'].replace({
    'M': 'Male',
    'F': 'Female',
    'MALE': 'Male',
    'FEMALE': 'Female'
})
```

```
In [ ]: data['Gender'].fillna(data['Gender'].mode()[0], inplace=True)
```

```
In [18]: data.isnull().sum()
```

```
Out[18]: Customer_ID      0
        Name              0
        Gender            0
        Age               950
        City             1014
        Signup_Date       0
        Last_Purchase_Date 1012
        Purchase_Amount   1020
        Feedback_Score    1022
        Email             0
        Phone_Number       0
        Country           731
        dtype: int64
```

Clean Age Column

```
In [19]: data['Age'].head(5)
```

```
Out[19]: 0      52.0
        1    51.0 years
        2      62.0
        3      40.0
        4      41.0
        Name: Age, dtype: object
```

```
In [20]: data['Age'] = data['Age'].astype(str)
        data['Age'] = data['Age'].str.extract('(\d+)')
```

```
In [21]: data['Age'] = pd.to_numeric(data['Age'])
```

```
In [22]: # fill missing ages
        data['Age'] = data['Age'].fillna(data['Age'].median())
```

Clean City Column

```
In [23]: data['City'].unique()
```

```
Out[23]: array([' KOLKATA ', nan, ' hyderabad ', ' CHENNAI ', ' kolkata ',
        ' BANGALORE ', ' Hyderabad ', ' HYDERABAD ', ' Mumbai ',
        ' chennai ', ' Delhi ', ' bangalore ', ' MUMBAI ', ' Bangalore ',
        ' Kolkata ', ' delhi ', ' Chennai ', ' mumbai ', ' DELHI '],
        dtype=object)
```

```
In [24]: data['City'] = data['City'].str.strip().str.title() # strip removes extra space
```

```
In [25]: data['City'] = data['City'].fillna(data['City'].mode()[0])
```

Clean Country Column

```
In [26]: data['Country'].unique()
```

```
Out[26]: array(['India', 'india', nan, 'InDia', 'IND'], dtype=object)
```

```
In [27]: data['Country'] = data['Country'].str.strip().str.upper()
```

```
In [28]: data['Country'] = data['Country'].replace({  
        'IND': 'India'  
    })
```

```
In [29]: data['Country'] = data['Country'].fillna('India')
```

Convert Date Columns

```
In [30]: data['Signup_Date'] = pd.to_datetime(data['Signup_Date'])
```

```
In [31]: data['Last_Purchase_Date'] = pd.to_datetime(  
        data['Last_Purchase_Date'])
```

```
In [32]: data['Last_Purchase_Date'] = data['Last_Purchase_Date'].fillna(  
        data['Last_Purchase_Date'].median()  
    )
```

Clean Purchase Amount

```
In [33]: data['Purchase_Amount'].isnull().sum()
```

```
Out[33]: np.int64(1020)
```

```
In [34]: data['Purchase_Amount'] = data['Purchase_Amount'].fillna(  
        data['Purchase_Amount'].median()  
    )
```

```
In [35]: data['Purchase_Amount'].isnull().sum()
```

```
Out[35]: np.int64(0)
```

Clean Feedback Score

```
In [36]: data['Feedback_Score'] = data['Feedback_Score'].fillna(  
        data['Feedback_Score'].median()  
    )
```

```
In [37]: data['Feedback_Score'].isnull().sum()
```

```
Out[37]: np.int64(0)
```

```
In [38]: data['Feedback_Score'] = data['Feedback_Score'].astype(int)
```

Verify Cleaned Data

```
In [39]: print(data.info())
```

```

<class 'pandas.core.frame.DataFrame'>
Index: 10185 entries, 0 to 10199
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Customer_ID           10185 non-null  object
 1   Name                  10185 non-null  object
 2   Gender                10185 non-null  object
 3   Age                   10185 non-null  float64
 4   City                  10185 non-null  object
 5   Signup_Date           10185 non-null  datetime64[ns]
 6   Last_Purchase_Date    10185 non-null  datetime64[ns]
 7   Purchase_Amount       10185 non-null  float64
 8   Feedback_Score        10185 non-null  int64
 9   Email                 10185 non-null  object
10   Phone_Number          10185 non-null  int64
11   Country               10185 non-null  object
dtypes: datetime64[ns](2), float64(2), int64(2), object(6)
memory usage: 1.0+ MB
None

```

```
In [40]: print(data.isnull().sum())
```

```

Customer_ID      0
Name              0
Gender           0
Age              0
City             0
Signup_Date      0
Last_Purchase_Date 0
Purchase_Amount  0
Feedback_Score   0
Email            0
Phone_Number     0
Country          0
dtype: int64

```

```
In [41]: print(data.head())
```

```

   Customer_ID      Name  Gender  Age  City  Signup_Date \
0  CUST4371      Paul Wilson   Male  52.0  Kolkata  2025-06-26
1  CUST5957    Jason Thomas   Male  51.0  Mumbai  2021-02-17
2  CUST3754  Brittney Martinez  Female  62.0  Hyderabad  2023-11-05
3  CUST2934    Brenda Pierce  Female  40.0  Hyderabad  2022-03-13
4  CUST5683  Matthew Carroll  Female  41.0  Chennai  2024-04-05

   Last_Purchase_Date  Purchase_Amount  Feedback_Score \
0      2025-05-17      26944.0          1
1      2025-07-22      44152.0          2
2      2024-12-08      31745.0          2
3      2025-10-02      39674.0          1
4      2024-12-15      24330.0          8

   Email  Phone_Number  Country
0  peckvictoria@example.com  2131107701  INDIA
1  owensanthony@example.com  1080761560  INDIA
2  tara39@example.org  8981006345  INDIA
3  berrynancy@example.org  8228064204  INDIA
4  denise84@example.org  2665569480  INDIA

```

Save Cleaned Dataset

```
In [44]: data.to_csv("cleaned_customer_sales_data.csv", index=False)

print("Dataset Saved Successfully")
```

Dataset Saved Successfully

```
In [45]: import os
print(os.getcwd())
```

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